# Contested Strategic Allocation: A Unified Framework for Games, Decision Problems, and the Founding of Orders

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**Abstract**

We introduce *Contested Strategic Allocation* (CSA), a mathematical framework that treats games, markets, bargaining, mechanism design, and single-agent planning as instances of a single structure: the strategic allocation of scarce resources under contention. CSA makes explicit five objects that standard game forms typically bundle into abstract action sets—resources, players, rules, allocation mappings, and payoffs—and it introduces *struggle* as the process through which contention unfolds. We define key terms first in ordinary language and then formally, so that the framework is intelligible before it is technical. We prove a decomposition theorem showing when a CSA problem separates into independent subproblems, characterize the structure of contention and the flow of struggle as graphs, and show that finite normal-form games, extensive-form games, exchange economies, cooperative games, mechanism design problems, and

Markov decision processes are all CSA instances. We then identify the *problem of first allocation*—the founding of an order that no prior rule allocates—as the framework's honest limit, and show that every founding occurs in one of four modes: force, agreement, convention, or accident, each carrying a characteristic legitimacy deficit. The first allocation may also *rig* the game: it can systematically advantage the founder's coalition, and this initial advantage is self-reinforcing—within-rules reform cannot eliminate it, and de-rigging requires refounding, which carries its own legitimacy deficit. We show that a game with a *live* founder advantage can never be purely zero-sum: the founder advantage is the trace, in the within-rules game, of a nonzero-sum over-rules game. We then apply the framework to political alignment, deriving the left–right axis and the full space of political axes from the contestable parameters of the CSA problem. Applications span personal decision-making, politics, and business strategy. CSA adds no expressive power beyond extensive-form games, but it changes the primitive ontology in ways that clarify comparative statics, mechanism design, and the sources of strategic difficulty.

**Keywords:** allocation, contention, struggle, first allocation, founder, rigged foundation, initial advantage, zero-sum, political axis, left–right, game theory, mechanism design, decision theory, strategy

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# 1 Introduction

Game theory begins with abstract action sets and payoff functions. This is powerful but obscures something important: what players actually do in most games is *claim scarce resources*—time, money, attention, territory, authority, capacity, information, risk—and what makes games hard is that claims *contend*. A chess move allocates pieces and tempo. A bid allocates budget and signals information. A political coalition allocates authority. A personal decision allocates the only life you have across competing ends.

Standard game forms treat these as special interpretations of an abstract structure. We propose the reverse: treat allocation under contention as primitive, and derive games, markets, bargaining, and mechanism design as special cases. The resulting framework—Contested Strategic Allocation (CSA)—has three virtues.

(1) **Unification.** Single-agent decision problems, multi-agent games, cooperative games, market equilibria, and mechanism design problems are all CSA instances, distinguished by the presence of contention, coalition structure, and design variables.

(2) **Structural clarity.** Contention becomes an explicit object—a graph on players, indexed by resources—rather than an implicit feature of payoff functions. This makes it possible to ask, and answer, when a problem decomposes, where strategic difficulty comes from, and when mechanisms help.

(3) **Applicability.** CSA's vocabulary—shares, claims, contention, struggle, rules, founders, foundations, rigging, axes—is closer to how decisions are actually described in personal life, politics, and business than the vocabulary of action profiles and payoff matrices.

CSA is not a new solution concept and does not add expressive power beyond extensive-form games with rich action spaces. Its contribution is organizational and structural: it identifies which features of a situation generate strategic difficulty, and it provides a common language for problems usually treated separately. It also identifies, with equal precision, the one question it cannot answer—the problem of first allocation—and thereby locates where a complementary theory is needed. And it derives, from the same structure, the familiar axes of political alignment. A game with a live founder advantage is never purely zero-sum; the left and the right are structural responses to that fact.

## 1.1 Related work

CSA synthesizes several traditions. Classical game theory [19, 13] provides the equilibrium concepts. Cooperative game theory [17, 3] provides coalitional allocation and the core. Mechanism design [6, 12] provides the design of rules. Resource allocation theory in economics provides the treatment of scarcity and prices. Decision theory and Markov decision processes provide the single-agent case. CSA's distinctive move is to make *allocation* and *contention* the organizing primitives, to introduce *struggle* as the process linking them, and to treat *rules* as an explicit, sometimes contested, object.

The problem of first allocation (§6) synthesizes a separate tradition: Nozick's original acquisition [14], Buchanan and Tullock's constitutional choice [1], Rawls's original position [15], Hume's first convention [5], Schmitt's constituent power [16], Lewis on coordination [9], Sugden on spontaneous order [18], Burke on inheritance [2], and Kelsen's *Grundnorm* [8]. Naming the problem "first allocation" unifies these treatments. The theory of the *rigged foundation* (§6.3)

formalizes a theme present in each of these traditions but rarely stated in this form: the founder's advantage, encoded in the foundation, is self-reinforcing and structurally nonzero-sum.

The analysis of political axes (§7.2.1–§7.2.2) connects to the literature on political cleavages—Lipset and Rokkan on the emergence of party systems, Inglehart on materialist and postmaterialist values, and the more recent work on the cultural backlash and the populist–elitist axis. The framework's contribution is to derive these cleavages from a single structural source rather than treating them as historical accidents.

### 1.2 Contributions

(1) An ordinary-language glossary of CSA's key terms, presented before any formalism (§2).

(2) A formal definition of CSA problems, strategies, equilibria, and contention structures, including struggle effort and weighted struggle networks (§3).

(3) A taxonomy of contention—rival, cooperative, and coordinative—expressed in both ordinary and formal language (§3.4).

(4) Representation results: normal-form games, extensive-form games, exchange economies, cooperative games, mechanism design, and MDPs as CSA instances (§4).

(5) A decomposition theorem and contention/struggle diagnostics (§5).

(6) A formal treatment of the problem of first allocation, including a regress theorem, a taxonomy of founders, four modes of founding with their legitimacy deficits, and a theory of the *rigged foundation*—the initial advantage conferred by the first allocation on the founder's coalition, the conditions under which it is self-reinforcing, and the structural obstacles to de-rigging (§6).

(7) A theorem connecting founder advantage to the nonzero-sum structure of the over-rules game, and a theory of erosion, revolution, and anti-rigging (§6.4–§6.5).

(8) A derivation of political alignment from the contention structure, including the left–right axis (§7.2.1) and the full space of political axes (§7.2.2).

(9) Applications to personal decision-making, politics, and business (§7).

(10) A candid discussion of scope and limitations (§8).

## 2 Key Terms in Ordinary Language

Before any formalism, we define the framework's vocabulary in plain speech. Each term is given with its everyday meaning, a concrete example, and a note on what it will become formally. The formal definitions in §3 are then translations of these ordinary meanings into mathematical objects.

### 2.1 Share

A **share** is the portion of something valuable that a person receives, keeps, or controls. We also call it a **reward**, a **portion**, a **stake**, or simply **what someone gets**. A worker's wage is a share of the firm's revenue. A country's territory is a share of the earth's surface. A student's grade is a share of the available credit. A person's evening is a share of their waking hours.

*Formally:* a share becomes a point $x_i$ in an allocation space $\mathcal{X}$, the vector of everything player $i$ receives.

## 2.2 Claim

A **claim** is what a person asserts, demands, proposes, or attempts in order to obtain, keep, or increase a share. A bid is a claim on a good. A move is a claim on a position. A budget request is a claim on funds. A request for someone's attention is a claim on their time. Claims may be compatible (two claims that can both be satisfied), incompatible (only one can be satisfied), or partially compatible (satisfying one reduces but does not eliminate the other).

*Formally:* a claim becomes an element $a_i \in A_i$, the set of actions available to player $i$.

## 2.3 Contention

**Contention** is the condition in which two or more claims bear on the same share—so that satisfying one affects the satisfaction of another. Two people contending for a promotion, two firms contending for a market, two goals contending for the same hour: in each case, contention is the *structural coupling* of claims through a shared share. Contention is not yet a fight; it is the situation that makes a fight possible. Some contention is **rival** (one person's share comes at another's

expense), some is **cooperative** (each person's share grows when the other's does), and some is **coordinative** (both want to agree on a joint allocation but care mainly about alignment).

*Formally:* contention becomes a family of labeled graphs $G_r$, one per resource, recording who is coupled with whom and how.

## 2.4 Struggle

A **struggle** is the activity of contending: the effort, time, resources, and risk that people expend in trying to secure, defend, or expand their shares. Contention is the *structure*; struggle is the *process*. People **struggle for** a share ("the struggle for independence," "the struggle for market share"), **struggle with** others against a common obstacle ("our shared struggle for recognition"), and **struggle over** which arrangement to adopt ("the struggle over the rules"). A struggle may be *latent* (contention exists but no one is expending effort), *active* (effort is being expended), or *resolved* (settled by rule, by exhaustion, or by exit).

*Formally:* struggle becomes effort $e_{ir} \geq 0$ allocated to resource $r$, with a budget constraint $\sum_r e_{ir} \leq B_i$, and the allocation mapping becomes effort-dependent.

## 2.5 Rules

**Rules** are the arrangements that determine how claims become shares: property rights, contracts, auctions, voting systems, constitutions, norms, and conventions. Rules may be given (a market's price mechanism), chosen (a firm's internal budgeting process), or contested (a constitutional convention, a standards war). Rules matter because they determine *which* struggles take place and *how* shares are assigned once claims are made. Much of politics and business strategy is not struggle within the rules but struggle over the rules.

*Formally:* rules become an object $M$, which may be exogenous, designed, or itself a claim.

## 2.6 Founder and foundation

A **foundation** is the settlement at the bottom of an order—the rules that are not themselves settled by any prior rule of that order. A **founder** is whoever or whatever lays the foundation. The founder may be an individual (a conqueror, a constitution-writer, a company founder, a parent), a collective (a constituent assembly, a founding partnership, a household), a process (a community's gradual convergence on a practice), or no one at all (history, drift, accident). The founder answers *who decided the rules*; the *mode of founding* answers *how*. A foundation is **rigged** when it systematically advantages the founder's coalition—when the initial advantage it

confers tracks who laid it rather than any founder-independent criterion. The initial advantage is **self-reinforcing** when the beneficiary coalition holds procedural rights sufficient to block reforms that would reduce it.

*Formally:* the foundation is the terminus of the regress of rule structures (§6); the founder and the mode of founding are classified in Definitions 6.3–6.4; initial advantage, rigging, and self-reinforcement are formalized in Definitions 6.6–6.11.

### 2.7 Player, resource, allocation

A **player** is anyone—or anything—that makes claims: a person, a firm, a party, a coalition, a government, a future self. A **resource** is anything scarce that can be claimed: time, money, attention, authority, capacity, information, risk, legitimacy. An **allocation** is the full assignment of shares to players. An allocation is **efficient** if no one can be made better off without making someone else worse off; it is **stable** if no player or group of players can profitably upset it.

*Formally:* players become $\mathcal{N}$, resources $\mathcal{R}$, allocations $\mathcal{X}$, efficiency Pareto-optimality, and stability equilibrium.

### 2.8 Strategy and equilibrium

A **strategy** is a plan for making claims—not just one claim, but a rule for what to claim given what one knows and what others have done. A strategy is thus itself an allocation: an allocation of claims across contingencies. An **equilibrium** is a profile of strategies such that no player (or coalition) can do better by changing their strategy unilaterally (or jointly, for coalitions). An equilibrium is a *settled outcome*: a way the struggle comes to rest when everyone is responding optimally to everyone else.

*Formally:* strategies become measurable maps $\sigma_i : I_i \to \Delta(A_i \times F_i)$; equilibrium becomes the appropriate fixed-point condition.

### 2.9 Axis

An **axis** is a stable disagreement about a parameter of the allocation problem: about what is divided, who is in the game, how much the rules cover, whether the founding is justified, or how the order should change. Axes are not arbitrary bundles of opinion; they are the political expression of structural positions in the contention graph. The left–right axis is the oldest and most stable, but it is one of five families.

*Formally:* axes are classified in Propositions 7.2 and 7.5.

### 2.10 The core picture

With these terms in hand, the framework's central claim can be stated in one ordinary sentence:

| A game is a struggle over shares, under rules, where what each player gets depends on what everyone claims. |
|---|

The formalism in §3 is a careful translation of that sentence into mathematics. The question of where the rules themselves come from—the problem of first allocation—is taken up in §6. The application of the framework to political alignment is taken up in §7.2.1.

## 3 The Framework

### 3.1 Primitive objects

**Definition 3.1** (CSA problem)**.** A *contested strategic allocation problem* is a tuple

$$\mathcal{G} = \big(\mathcal{N},\, \mathcal{R},\, \Theta,\, \{A_i\}_{i\in\mathcal{N}},\, \{I_i\}_{i\in\mathcal{N}},\, \{B_i\}_{i\in\mathcal{N}},\, \Phi,\, M,\, \{u_i\}_{i\in\mathcal{N}},\, T\big)$$

with the following interpretation.

- $\mathcal{N} = \{1, \ldots, n\}$ is a finite set of **players** (agents, coalitions, goals, or future selves).
- $\mathcal{R}$ is a finite set of **resource types** (time, money, attention, authority, capacity, information, risk, ...).
- $\Theta$ is a set of **states** (preference parameters, shocks, private information).
- $A_i$ is player $i$'s **claim set**: the set of allocations of resources, effort, or moves that $i$ can propose, commit to, or execute.
- $I_i$ is $i$'s **information partition** over $\Theta \times A_{-i}$: what $i$ knows when choosing.
- $B_i \geq 0$ is $i$'s **struggle budget**: the total effort available for contending across resources. Let

$$F_i \;=\; \Big\{ e_i \in \mathbb{R}_+^{|\mathcal{R}|} : \textstyle\sum_{r \in \mathcal{R}} e_{ir} \leq B_i \Big\}$$

  be $i$'s **effort set**, and $F = \prod_{i \in \mathcal{N}} F_i$.
- Write $A = \prod_{i \in \mathcal{N}} A_i$, and let $\mathcal{X} \subseteq \prod_{r \in \mathcal{R}} \mathbb{R}_+^n$ be the **allocation space**. The **allocation mapping**

$$\Phi : A \times F \times \Theta \times M \;\longrightarrow\; \mathcal{X}$$

  determines how joint claims, struggle efforts, states, and rules produce shares of each resource for each player.
- $M$ is the **rule structure**: property rights, contracts, auctions, voting rules, norms, enforcement technologies. $M$ may be exogenous, designed, or contested; its origin is the subject of §6.
- $u_i : \mathcal{X} \times A \times F \times \Theta \to \mathbb{R}$ is player $i$'s **payoff**, which includes the value of $i$'s share and the cost of $i$'s struggle.
- $T$ is a **transition map** $s_{t+1} = T(s_t, a_t, x_t)$, defining the dynamic version; the static case has $T \equiv \varnothing$.

The key structural choice is that *allocation* $x \in \mathcal{X}$ is separated from *claims* $a \in A$, from *struggle* $e \in F$, and from *rules* $M$. In a standard normal-form game these are collapsed: actions directly determine payoffs. In CSA, claims and efforts determine allocations *through* $\Phi$, and allocations determine payoffs *given* the rules. This separation is what allows the framework to accommodate markets (where $\Phi$ is a price mechanism), bargaining (where $\Phi$ is an agreed split), and planning (where $\Phi$ is the identity, $n = 1$).

### 3.2 Strategies and outcomes

**Definition 3.2** (Strategy)**.** A *strategy* for player $i$ is a measurable map

$$\sigma_i : I_i \to \Delta(A_i \times F_i),$$

assigning to each information set a distribution over claims and effort vectors. A *strategy profile* is $\sigma = (\sigma_1, \ldots, \sigma_n)$.

**Definition 3.3** (Induced allocation and payoff)**.** Given a strategy profile $\sigma$, a state $\theta$, and rules $M$, the induced allocation is

$$x(\sigma, \theta, M) \;=\; \mathbb{E}_{(a,e) \sim \sigma}\big[\Phi(a, e, \theta, M)\big],$$

and $i$'s expected payoff is

$$U_i(\sigma, \theta, M) \;=\; \mathbb{E}_{(a,e) \sim \sigma}\big[u_i(x(\sigma, \theta, M), a, e, \theta)\big].$$

### 3.3 Equilibrium

**Definition 3.4** (Nash equilibrium)**.** Let $\theta \in \Theta$ be fixed (common knowledge). A profile $\sigma^*$ is a *Nash equilibrium* of $\mathcal{G}$ if for all $i$ and all $\sigma_i'$,

$$U_i(\sigma^*, \theta, M) \;\geq\; U_i(\sigma_i', \sigma_{-i}^*, \theta, M).$$

When $\theta$ is stochastic and not commonly observed, see Definition 3.6.

**Definition 3.5** (Coalitional equilibrium)**.** A profile $\sigma^*$ is *coalition-proof* if no coalition $S \subseteq \mathcal{N}$ and deviation $\sigma_S'$ satisfies

$$U_i(\sigma_S', \sigma_{-S}^*) > U_i(\sigma^*) \quad \text{for all } i \in S.$$

The core, the Shapley value, and cooperative solution concepts can all be expressed as coalitional equilibria of CSA problems in which $\Phi$ is a surplus-splitting rule (§4.4).

**Definition 3.6** (Bayesian equilibrium)**.** When $\Theta$ is stochastic and the information partitions $I_i$ are non-trivial, replace $U_i$ with the conditional expectation $\mathbb{E}[U_i(\sigma, \theta, M) \mid h]$ given $i$'s information set $h \in I_i$, and require optimality at every $h$.

### 3.4 Contention structure and struggle network

Contention is the structural feature that makes a CSA problem *strategic* rather than merely *allocative*. Struggle is the activity through which contention is pursued.

**Definition 3.7** (Rival contention)**.** Players $i$ and $j$ *rival-contend* over resource $r$ if, holding all else fixed,

$$\frac{\partial u_i}{\partial x_{jr}} < 0 \quad \text{and} \quad \frac{\partial u_j}{\partial x_{ir}} < 0.$$

Each strictly prefers the other to receive less of $r$.

**Definition 3.8** (Cooperative contention)**.** Players $i$ and $j$ *cooperative-contend* over $r$ if

$$\frac{\partial u_i}{\partial x_{jr}} > 0 \quad \text{and} \quad \frac{\partial u_j}{\partial x_{ir}} > 0.$$

Each prefers the other to receive more of $r$—partnership, team production, alliance, or complementarity.

**Definition 3.9** (Coordinative contention)**.** Players $i$ and $j$ *coordinate* over $r$ if both prefer a specific joint allocation over misaligned ones, but neither prefers the other's share per se—e.g., driving on the same side of the road, or adopting a common standard.

**Definition 3.10** (Contention structure)**.** The *contention structure* of $\mathcal{G}$ is a family of labeled graphs $\{G_r^{\text{riv}}, G_r^{\text{coop}}, G_r^{\text{coord}}\}_{r \in \mathcal{R}}$ with

$$(i,j) \in G_r^{\text{riv}} \iff i, j \text{ rival-contend over } r,$$

and similarly for the cooperative and coordinative graphs. The *aggregate contention graph* is

$$G \;=\; \bigcup_{r \in \mathcal{R}} \big(G_r^{\text{riv}} \cup G_r^{\text{coop}} \cup G_r^{\text{coord}}\big).$$

**Definition 3.11** (Struggle network)**.** The *weighted struggle network* is the family of weighted graphs $\{\tilde{G}_r\}_{r\in\mathcal{R}}$ with edge weights

$$w_{ijr} \;=\; \frac{\partial^2 u_i}{\partial e_{ir}\,\partial e_{jr}},$$

the marginal effect on $i$'s payoff of $j$'s struggle effort over resource $r$. Positive weights indicate cooperative struggle (efforts reinforce), negative weights indicate rival struggle (efforts offset). An edge with $|w_{ijr}|$ below a threshold is *latent*: the contention exists structurally but no significant struggle is being expended.

**Definition 3.12** (Activity)**.** Contention over resource $r$ is *active* if $\sum_i e_{ir} > 0$; *latent* if the structural edges exist but $\sum_i e_{ir} = 0$; and *resolved* if the rule structure $M$ assigns shares without residual struggle.

This taxonomy drives the decomposition results in §5. The ordinary-language gloss is: **rival contention produces struggle against; cooperative contention produces struggle with; coordinative contention produces struggle over.** The grammar of struggle encodes the structure of contention.

# 4 Special Cases

We now show that the standard models of decision and game theory are CSA instances. In each case we identify shares, claims, rules, struggle, and the allocation mapping.

## 4.1 Normal-form games

**Proposition 4.1.** *Every finite normal-form game $G = (\mathcal{N}, A, g)$ is a CSA problem.*

*Proof.* Set $\mathcal{R} = \{1\}$ (a single "payoff resource"), $\mathcal{X} = \mathbb{R}^n$ with $x_i$ the share of player $i$, $M = \varnothing$, $\Theta$ a singleton, $I_i = A_{-i}$ (or trivial), $B_i = 0$ so that $F_i = \{0\}$ (no explicit struggle), $\Phi(a) = g(a)$, and $u_i(x, a) = x_i$. Then $U_i(\sigma) = \mathbb{E}[g_i(a)]$, and Nash equilibria coincide. Contention structure is read off the sign of $\partial g_i/\partial a_j$ where differentiable. □

The representation makes explicit what standard normal-form games hide: that payoffs *are* shares, and that strategic interaction is precisely contention over the payoff resource.

## 4.2 Extensive-form games

**Proposition 4.2.** *Every finite extensive-form game with perfect recall is a CSA problem with $T \neq \varnothing$.*

*Sketch.* Take $\mathcal{R}$ to include "information" and "tempo" alongside material payoffs; $A_i$ are behavioral strategies; $M$ encodes the game tree (chance, move order, information sets); $T$ encodes the transition from one node to the next; $\Phi$ accumulates terminal shares. By Kuhn's theorem, mixed and behavioral strategies are outcome-equivalent under perfect recall, so the mixed equilibria of the CSA correspond exactly to those of the original extensive-form game. □

The CSA representation foregrounds what extensive-form games typically background: that moving first, revealing information, or committing to a sequence is an *allocation of tempo and information*—shares in their own right.

### 4.3 Exchange economies

**Proposition 4.3.** *Every competitive exchange economy with $n$ agents and $m$ goods is a CSA problem in which $\Phi$ is the Walrasian allocation mapping.*

*Sketch.* Let $\mathcal{R}$ be the set of goods, $A_i = \mathbb{R}^m_+$ be $i$'s consumption claim, $M$ be the price system (possibly determined endogenously), and $\Phi(a, M)$ solve $\sum_i a_i = \sum_i \omega_i$ with each $a_i$ maximizing $u_i$ at prices $p(M)$. Walrasian equilibrium corresponds to a *competitive equilibrium* of the CSA: each player's claim maximizes

$u_i$ taking prices as given, and markets clear. This coincides with the Nash equilibrium of the CSA in the large-market limit, or under a mechanism $M$ that constrains players to be price-takers. The contention structure includes the complete rival graph on each good. □

### 4.4 Cooperative games and bargaining

**Proposition 4.4.** *Every transferable-utility cooperative game $(\mathcal{N}, v)$ is a CSA problem whose coalitional equilibria are the core.*

*Sketch.* Let $\mathcal{R} = \{1\}$ be a single divisible surplus, $A_i = \mathbb{R}_+$ be $i$'s claim, $\Phi(a, M)$ be a surplus-splitting rule with $M$ encoding the characteristic function $v$, and $u_i(x) = x_i$. A profile is coalition-proof iff no coalition $S$ can deviate to an allocation exceeding $v(S)$—precisely the core condition. □

### 4.5 Mechanism design

**Proposition 4.5.** *Every mechanism design problem is a CSA problem with an additional* designer *player whose claim set is the rule space $M$.*

*Sketch.* Add player 0 (the designer) with $A_0 = M$, $u_0$ the designer's objective (welfare, revenue, efficiency), and let the other players' strategies be conditioned on $M$. The designer's strategy is a *rule allocation*; the equilibrium concept is the appropriate solution concept for the induced game. The revelation principle becomes the statement that any CSA problem with arbitrary claim sets $(A_i)$ and any equilibrium can be replaced by an outcome-equivalent CSA problem with direct claims $A_i = \Theta_i$ and a truthful equilibrium. □

### 4.6 Single-agent decision problems

**Proposition 4.6.** *Every finite-horizon Markov decision process is a CSA problem with $|\mathcal{N}| = 1$.*

*Sketch.* Let $\mathcal{R} = \{\text{reward}, \text{cost}\}$, $\mathcal{X} = \mathbb{R}^2_+$, $A_1$ the action set, and $\Phi(a, e, \theta) = (r(a, \theta), c(a, \theta))$. Let $F_1$ be trivial (no contention, hence no struggle), $T$ the MDP transition, and $u_1$ the (possibly discounted) reward function. A policy is a strategy. The structural contention graph is empty, so (as shown in §5) the problem decomposes into an independent single-agent decision problem—not strategically trivial, since uncertainty and dynamics remain, but free of contention. □

This is worth emphasizing: **contention is what separates games from decision problems, not the number of players alone.** A single agent facing internal conflict—multiple goals, future selves, self-control—can have a non-trivial contention structure even with $n = 1$. We return to this in §7.1.

## 5 Contention, Struggle, and Decomposition

We now show that the contention and struggle structures determine when a CSA problem separates into independent subproblems, and what mechanisms can do when it does not.

### 5.1 Strategic separability

For a set $S \subseteq \mathcal{N}$ and vectors $x \in \mathcal{X}$, $a \in A$, $e \in F$, write $x|_S$, $a|_S$, $e|_S$ for the restrictions to players in $S$.

**Definition 5.1** (Strategic separability)**.** Let $\{\mathcal{N}_1, \ldots, \mathcal{N}_k\}$ be a partition of $\mathcal{N}$. A CSA problem $\mathcal{G}$ is *strategically separable* across the partition if:

(1) (Allocation separability) $\Phi(a, e, \theta, M) = \big(\Phi_1(a|_{\mathcal{N}_1}, e|_{\mathcal{N}_1}, \theta, M), \ldots, \Phi_k(a|_{\mathcal{N}_k}, e|_{\mathcal{N}_k}, \theta, M)\big)$; and

(2) (Payoff separability) For each $m$ and each $i \in \mathcal{N}_m$, $u_i$ depends on $x$, $a$, and $e$ only through $(x|_{\mathcal{N}_m}, a|_{\mathcal{N}_m}, e|_{\mathcal{N}_m})$.

Let $\mathcal{G}_m$ denote the CSA restricted to $\mathcal{N}_m$ with the induced allocation mapping and payoffs.

**Theorem 5.2** (Decomposition)**.** *If $\mathcal{G}$ is strategically separable across $\{\mathcal{N}_m\}$, then*

$$\mathrm{NE}(\mathcal{G}) \;=\; \prod_{m=1}^{k} \mathrm{NE}(\mathcal{G}_m).$$

*Proof.* Let $\sigma^* \in \mathrm{NE}(\mathcal{G})$. Fix $m$ and $i \in \mathcal{N}_m$. Any deviation $\sigma_i'$ affects $U_i$ only through $\mathcal{G}_m$, by payoff separability. Since $\sigma^* \in \mathrm{NE}(\mathcal{G})$, no such deviation is profitable; hence $\sigma^*|_{\mathcal{N}_m} \in \mathrm{NE}(\mathcal{G}_m)$. Conversely, if $\sigma_m^* \in \mathrm{NE}(\mathcal{G}_m)$ for each $m$, then any deviation by $i \in \mathcal{N}_m$ is a deviation in $\mathcal{G}_m$ and hence unprofitable. □

**Corollary 5.3.** *If the* active *struggle network is empty—contention exists structurally but no effort is being expended—then $\mathcal{G}$ decomposes into independent single-agent problems or pure coordination problems, and equilibrium is unique up to individual indifference.*

**Corollary 5.4.** *If the active struggle network is disconnected with components $C_1, \ldots, C_k$, and if strategic separability holds across the components, then equilibria decompose across components.*

**Corollary 5.5.** *If contention is latent (structural but inactive), the allocation is determined by rules $M$ alone; if contention is active, the allocation is determined jointly by rules and struggle; if contention is resolved, the*
*allocation is determined by rules and the resolution mechanism.*

The theorem is elementary, but it is *the* structural result the CSA framework is designed to express: **strategic difficulty is localized to the connected components of the active struggle network.** In a large CSA problem—a firm, a polity, a personal life—one should first ask which resources actually generate active struggle, and among whom. Everything outside that network is a decision problem or a coordination problem, not a strategic one.

### 5.2 Contention and struggle indices

For diagnostic purposes, define the *rival contention index*

$$\kappa(\mathcal{G}) \;=\; \frac{\big|\{(i,j,r) : i \neq j,\ (i,j) \in G_r^{\mathrm{riv}}\}\big|}{n(n-1)\,|\mathcal{R}|} \;\in\; [0,1],$$

measuring the density of rival contention, and the *struggle intensity*

$$\sigma(\mathcal{G}) \;=\; \frac{\sum_{i,r} e_{ir}}{\sum_i B_i} \;\in\; [0,1],$$

measuring the fraction of available effort being expended. These are not theorem-generating quantities but useful comparative diagnostics. In families of CSA problems parameterized by $\kappa$ and $\sigma$:

- As $\kappa \to 0$, equilibria typically converge to independent optima and coordination outcomes.
- As $\sigma \to 0$, the situation reduces to allocation by rule.
- As $\kappa \to 1$ and $\sigma \to 1$, equilibrium allocations diverge from Pareto-efficient allocations whenever externalities are not internalized (Proposition 5.6).
- The value of a mechanism—the welfare gain from choosing $M$ optimally—is increasing in $\kappa\sigma$ in standard quasi-linear environments (e.g., [12]), though the exact relationship is environment-dependent.

**Proposition 5.6** (Contention, struggle, and inefficiency)**.** *Suppose $\mathcal{G}$ has rival contention (some $G_r^{\text{riv}} \neq \varnothing$) and $M = \varnothing$. Then generically the Nash equilibrium allocation is not Pareto-efficient.*

*Sketch.* Rival contention means each player's share exerts a negative externality on another. In the absence of a rule internalizing the externality, the first-order conditions for individual optimality differ from the conditions for Pareto optimality. Genericity follows because the set of payoff functions for which the two coincide despite rivalry has measure zero. □

**Proposition 5.7** (Mechanisms as struggle regulation)**.** *For any CSA problem $\mathcal{G}$ with rival contention and $M = \varnothing$, if there exists a rule $M^*$ (Pigouvian prices, Coasian property rights, VCG transfers in quasi-linear environments, or a negotiated contract) such that the induced equilibrium allocation is Pareto-efficient, then $M^*$ is a* struggle-regulating *mechanism.*

*Remark* 5.8. The proposition is a template, not a deep result: it packages standard mechanism-design results in CSA language. Its value is to make the *object* of design explicit—$M^*$ regulates contention by changing the allocation mapping $\Phi(\cdot, \cdot, \cdot, M)$, not by changing preferences or by eliminating struggle. The essence of institutional design is to channel struggle into productive contention and away from destructive contention.

# 6 The Problem of First Allocation: Founding and Rigging

The framework of §3–§5 takes the rule structure $M$ as given—exogenous, designed, or contested—and derives allocations, equilibria, and decomposition results from it. This section confronts the question the framework brackets: *where does $M$ come from?* The answer has two parts. First, the regress of rules terminates in a *foundation* laid by a *founder* in one of four modes. Second, the foundation is itself an *allocation*—of rule-making authority, and through it, of every subsequent share. It can systematically advantage some players over others. We call this the *rigged foundation*, and we show that its advantage is self-reinforcing and structurally nonzero-sum.

## 6.1 The regress and its terminus

**Definition 6.1** (First allocation)**.** The *first allocation* of a CSA problem is the allocation of rule-making authority $M$ itself—the allocation that no prior rule of the problem allocates.

If $M_0$ is allocated by a rule structure $M_1$, then $M_1$ must be allocated by $M_2$, and so on:

$$M_0 \leftarrow M_1 \leftarrow M_2 \leftarrow \cdots$$

Either the sequence is infinite—rules all the way down, with no foundation—or it terminates. It terminates at the *foundation of the order*: the settlement that is not itself settled by any prior rule of that order.

**Proposition 6.2** (Regress and foundation)**.** *Every CSA problem either admits an infinite regress of rule structures or terminates in a foundation of the order that is not itself rule-allocated.*

*Proof.* The sequence $\{M_k\}$ is either infinite or finite. If finite, its last element is by definition not allocated by any rule of the problem; it is the foundation. □

The foundation is the *constitutive outside* of the framework. It can be described but not derived. It appears in the literature under many names: as the state of nature or original acquisition in political philosophy [14, 4, 11], as the original position in normative theory [15], as constituent power in

constitutional theory [16], as the constitutional choice in public choice [1], as the first convention in the Scottish Enlightenment [5], as the *Grundnorm* in legal theory [8], as the initial endowment in welfare economics, and as the disagreement point in bargaining theory [13]. Naming the problem "first allocation" unifies these treatments: they are the same structural moment in different registers.

## 6.2 The founder and the modes of founding

The foundation is laid by a *founder*—the agent or process that establishes the first rules.

**Definition 6.3** (Founder)**.** A *founder* of a CSA problem is whatever establishes the first rules of the order: an individual (a conqueror, a constitution-writer, a company founder, a parent), a collective (a constituent assembly, a founding partnership, a household), a process (a community's gradual convergence on a practice), or no one at all (history, drift, accident).

The founder answers *who decided the rules*; the mode of founding answers *how*.

**Definition 6.4** (Modes of founding)**.** The first rules of an order are established in one of four modes:

(i) **Force.** The founder imposes the rules by seizure, conquest, coup, or fiat [16].

(ii) **Agreement.** The founder or founders consent to the rules by compact, constitution, charter, or vow, actual or hypothetical [4, 11, 15, 1].

(iii) **Convention.** The rules emerge from convergent practice without explicit agreement or a single decision [5, 9, 18].

(iv) **Accident.** The rules are the residue of history—path dependence, inheritance, drift [2].

The modes are exhaustive: every founding is either *deliberate* (force, agreement) or *emergent* (convention, accident), and either *concentrated* in an agent (force) or *diffuse* (agreement, convention, accident). In the force and agreement modes the founder is identifiable; in the convention mode the founder is a community over time; in the accident mode there is no founder at all.

Each mode carries a characteristic *legitimacy deficit*:

| Mode | Founder | Legitimacy deficit |
|---|---|---|
| Force | Imposing agent | Not legitimate by the rules it founds |
| Agreement | Consenting collective | Hypothetical or incomplete |
| Convention | Community over time | Not chosen |
| Accident | No one | Not justified |

**Proposition 6.5** (Legitimacy deficit)**.** *No founding can fully justify itself by the standards of the order it founds.*

*Sketch.* For each mode, the deficit is structural. Force is by definition the imposition of rules without prior authorization, so it cannot be legitimate under the rules it establishes. Agreement is either actual—in which case it is incomplete, since not all affected parties can be present at the founding—or hypothetical, in which case it is not an actual authorization. Convention emerges without decision, so it cannot be chosen. Accident involves no justifying agent at all. □

### 6.3 The rigged foundation: initial advantage

A foundation is not merely a terminus of the regress. It is also an *allocation*—of rule-making authority, and through it, of every subsequent share. This section treats the foundation as a strategic object: something that can favor some players over others, and whose favoritism persists.

**Definition 6.6** (Initial advantage)**.** Let $\mathcal{G}$ be a CSA problem with foundation $M^*$, and let $M^0$ be a *benchmark* rule structure (for instance, symmetric rules, equal treatment, or the rules that would obtain under a veil of ignorance). The *initial advantage* conferred by $M^*$ on player $i$, relative to $M^0$, is

$$\alpha_i(M^*, M^0) \;=\; U_i(\sigma^*(M^*), M^*) \;-\; U_i(\sigma^*(M^0), M^0),$$

where $\sigma^*(M)$ denotes an equilibrium of the CSA under rule structure $M$.

The vector $\alpha = (\alpha_1, \ldots, \alpha_n)$ measures how the foundation shifts each player's equilibrium share relative to the benchmark.

**Definition 6.7** (Rigged foundation)**.** A foundation $M^*$ is *rigged* relative to benchmark $M^0$ if there exists a coalition $S \subseteq \mathcal{N}$ such that

(i) $\alpha_i(M^*, M^0) > 0$ for all $i \in S$;

(ii) $\alpha_j(M^*, M^0) < 0$ for some $j \notin S$; and

(iii) the pattern of advantage tracks the identity of the founder (or the founder's coalition) rather than efficiency, desert, or any founder-independent criterion.

The coalition $S$ is the *beneficiary coalition* of the foundation.

Condition (iii) is the substantive one. A foundation may advantage some players for benign reasons—efficiency, coordination, the correction of prior injustice. It is *rigged* when the advantage is explained by *who laid it*, not by any reason that could be stated independently of the founder's interest.

*Remark* 6.8 (Equilibrium selection). The initial advantage $\alpha_i$ depends on the equilibrium selected under $M^*$ and under $M^0$. In games with multiple equilibria, $\alpha_i$ is not

uniquely defined, and the comparison of foundations may depend on equilibrium selection. This is a known problem in comparative institutional analysis and is not specific to CSA. When equilibria are unique, or when a common refinement selects the relevant equilibrium, $\alpha_i$ is well-defined.

*Remark* 6.9 (Rigging is not all-or-nothing). Rigging is a matter of degree. A foundation may be mildly rigged (small $\alpha$ concentrated in a small beneficiary coalition), severely rigged (large $\alpha$ in a large beneficiary coalition, with large negative $\alpha$ for others), or effectively neutral (small $\alpha$ across the board). The legitimacy deficit of Proposition 6.5 is amplified in proportion to the magnitude of the rigging: a severely rigged foundation is not merely unjustified in the abstract but unjustified in a way that is visible in every subsequent allocation.

**Definition 6.10** (Types of rigging)**.** Rigging takes five characteristic forms:

(i) **Explicit favoritism.** The rules directly privilege the founder's coalition: property qualifications for the franchise, founder shares with superior voting rights, inheritance rules that favor one line, constitutional entrenchment of a particular interest.

(ii) **Path dependence.** Early advantages compound: first-mover advantages, network effects, accumulated capital, incumbency. The foundation does not explicitly favor anyone, but its timing does.

(iii) **Information asymmetry.** The founder knows the rules better than later players, or encodes private information into the foundation. The foundation favors those who wrote it.

(iv) **Status-quo bias.** The foundation makes change costly, and the beneficiaries of the status quo are therefore advantaged relative to those who would prefer a different allocation.

(v) **Coordinative selection.** The foundation selects one convention among many equivalent ones, and the selected convention favors some players (language, standards, driving side, calendar). Usually mild, but potentially severe when the convention is tied to identity.

**Definition 6.11** (Self-reinforcing rigging)**.** A rigged foundation is *self-reinforcing* if the beneficiary coalition $S$ holds procedural rights under $M^*$—vetoes, supermajority requirements, agenda control, appointment power—sufficient to block reforms that would reduce $\alpha_i$ for $i \in S$.

Self-reinforcing rigging is the generic case in durable orders. A founder who can encode an advantage in the foundation can usually also encode the procedural protections that preserve it.

**Proposition 6.12** (Persistence of initial advantage)**.** *Let $M^*$ be a self-reinforcing rigged foundation in favor of coalition $S$. Then no within-rules reform $M'$ can reduce $\alpha_i(M', M^0)$ below zero for any $i \in S$ without $i$'s consent. Since $S$ has no incentive to consent to a reduction in its own advantage, the advantage persists.*

*Sketch.* By Definition 6.11, $S$ holds procedural rights under $M^*$ sufficient to block any reform that reduces $\alpha_i$ for

$i \in S$. A reform $M'$ that reduces $\alpha_i$ below zero for some $i \in S$ requires $i$'s consent (by the definition of $M'$ as within-rules and the procedural rights $S$ holds). If $i$'s consent is required, $i$ can block the reform. Since $i$'s advantage is precisely what $i$ would lose, $i$ has no incentive to consent. The advantage persists. □

**Proposition 6.13** (Reform constraint)**.** *Under the hypotheses of Proposition 6.12, any within-rules reform $M'$ that is Pareto-improving relative to $M^*$ must preserve the relative advantage of $S$: $\alpha_i(M', M^*) \geq 0$ for all $i \in S$.*

*Sketch.* A Pareto improvement makes every player weakly better off and at least one strictly better off. If the reform reduced $\alpha_i(M', M^*) < 0$ for some $i \in S$, then $i$ would be strictly worse off relative to $M^*$, and the reform would not be Pareto-improving. Hence $\alpha_i(M', M^*) \geq 0$ for all $i \in S$. □

**Proposition 6.14** (Refounding as de-rigging)**.** *Absent the consent of the beneficiary coalition, eliminating a rigged foundation requires refounding—a change in $M^*$ itself—which by Proposition 6.2 is not rule-governed and therefore carries its own legitimacy deficit.*

*Sketch.* By Proposition 6.12, within-rules reform cannot eliminate $S$'s advantage without $S$'s consent. Absent that consent, elimination requires a change in $M^*$, which is by definition a refounding. A refounding is not governed by the rules of $M^*$ (it replaces them), so it must be justified by something other than $M^*$. By Proposition 6.5, no founding can fully justify itself by the rules it founds. Hence the refounding carries its own legitimacy deficit. □

*Remark* 6.15 (The regress of de-rigging). Proposition 6.14 has a disquieting implication. Every attempt to correct a rigged foundation requires a new founding, and every new founding is itself rigged in some measure (if only by the interests of those who carry it out). The regress of rules is matched by a regress of *corrections*: each correction is a new founding, with its own beneficiaries and its own legitimacy deficit. There is no rule-governed path to a fully unrigged foundation. This is not a counsel of despair—refounding can genuinely improve on what it replaces—but it is a counsel of realism about the limits of reform.

*Remark* 6.16 (Rigging, contention, and decomposition). Rigging reshapes the contention structure of §3.4. In a rigged order, the beneficiary coalition contends less (the rules already favor it) and the disadvantaged contend more (they must struggle against the rules as well as within them). The aggregate contention graph is asymmetric: the disadvantaged have edges to the foundation itself, not merely to one another. This is why rigged orders often appear calm on the surface—the beneficiaries have little to struggle for—while the disadvantaged experience constant, low-grade struggle that the beneficiaries do not see. Rigging also reshapes decomposition: if the beneficiary coalition's advantage depends on bundling two otherwise separable issues—say, a popular policy with a rigged procedural rule—

then the rigging creates artificial coupling. The decomposition theorem of §5 then applies not to the underlying situation but to the rigged one: the issues appear strategically coupled because the rigging makes them so, and unbundling them is itself a de-rigging move.

### 6.4 Founder advantage and the nonzero-sum structure

A common intuition treats strategic situations as zero-sum: what one player gains, another loses, and the total is fixed. This subsection shows that a game with a *live* founder advantage—one whose foundation is contestable—can never be purely zero-sum. The founder advantage is the trace, in the within-rules game, of an over-rules game that is structurally nonzero-sum.

**Definition 6.17** (Zero-sum)**.** A CSA problem $\mathcal{G}$ is *zero-sum* if the total payoff $\sum_{i\in\mathcal{N}} u_i(x, a, e, \theta)$ is constant across all outcomes reachable under equilibrium play. It is *zero-sum within* $M$ if the total is constant across all plays consistent with fixed $M$, and *zero-sum over* $M$ if $\sum_i U_i(\sigma^*(M), M)$ is constant in $M$.

The distinction matters because a game can be zero-sum within a fixed foundation and nonzero-sum over foundations. A fixed budget divided by rigged rules is zero-sum within the rules; but the rules themselves can be changed, and changing them can change the total.

**Proposition 6.18** (Founder advantage and nonzero-sum structure)**.** *Let $\mathcal{G}$ be a CSA problem with foundations $M^*$ and $M^0$. Suppose $M^*$ is rigged relative to $M^0$ with nonzero advantage vector $\alpha$ (Definition 6.6), and suppose the foundation is contestable (de-rigging via refounding is possible; Proposition 6.14). Then $\mathcal{G}$, including the possibility of refounding, is not zero-sum over $M$.*

*Proof.* By Proposition 6.14, de-rigging requires refounding—a change in $M^*$. A refounding is a structure-setting move: it replaces the allocation mapping itself. If the total payoff $\sum_i U_i(\sigma^*(M), M)$ were constant across all $M$, then any refounding would be purely redistributive—it could only move shares around, never change their sum. But a de-rigging refounding raises the payoffs of the disadvantaged coalition $j \notin S$, and by Proposition 6.12 the beneficiaries $i \in S$ cannot be made worse off without their consent. So a de-rigging refounding raises the total (or at least does not lower it while raising some payoffs). The total is therefore not constant across $M$. Hence the game over $M$ is nonzero-sum. □

**Corollary 6.19** (Zero-sum thinking)**.** *Zero-sum reasoning is reasoning within a fixed foundation. It is correct when the foundation is genuinely fixed and incorrect when it is not. The players most likely to insist that the game is zero-sum are those who benefit from the current foundation, because zero-sum framing keeps the struggle within the rules and prevents the over-rules game from being contested.*

**Corollary 6.20** (Refounding is nonzero-sum)**.** *A refounding—a change in $M^*$—is never a zero-sum move. It changes the total payoff, not merely its distribution. This is why refoundings are both dangerous and generative: they can create a better foundation or a worse one, and the uncertainty about which is why they are so contested.*

**Corollary 6.21** (Structural inequality)**.** *Inequality produced by a rigged foundation is not a zero-sum transfer. It is the trace of a nonzero-sum process: the founder's structural advantage compounds, and the compounding changes the total. This is uncomfortable for both redistributionist and meritocratic accounts: the redistributionist wants to say the advantage is a transfer (zero-sum), the meritocrat wants to say it is deserved (founder-independent). The framework says it is structural, and structural advantages are neither transfers nor deserts.*

*Remark* 6.22 (The slogan). A useful compression: *zero-sum is the arithmetic of players who cannot see the foundation.* The framework's insistence on the first allocation is, in part, an insistence that the arithmetic is usually incomplete.

## 6.5 Erosion, revolution, and anti-rigging

Proposition 6.12 establishes that a self-reinforcing rigged foundation resists within-rules reform. The persistence is conditional, however, on the beneficiary coalition retaining the procedural rights that make the rigging self-reinforcing. This subsection identifies the conditions under which that retention fails, and distinguishes three responses to rigging.

**Definition 6.23** (Erosion)**.** A rigged foundation *erodes* if, without refounding, the advantage $\alpha_i$ of some beneficiary $i \in S$ declines over time. Erosion occurs through one of three channels:

(i) **Coalition mortality.** The beneficiary coalition fragments, dies out, or is diluted, so that the procedural rights it holds are dispersed or lost.

(ii) **Countervailing leverage.** New players accumulate the resources on which the rules depend—capital, votes, information, labor, legitimacy—sufficient to contest the rigging from within.

(iii) **Reinterpretation.** The rules are read differently by later interpreters, so that the advantage they confer changes without the rules themselves changing.

**Definition 6.24** (De-rigging, re-rigging, anti-rigging)**.** A refounding $M'$ of a rigged foundation $M^*$ is:

(i) a *de-rigging* if it eliminates the advantage $\alpha_i$ for $i \in S$;

(ii) a *re-rigging* if it replaces one beneficiary coalition with another while preserving the structural features of rigging (Definition 6.10);

(iii) an *anti-rigging* if it changes the structure so that future rigging is harder—through sunset clauses, rotation, transparency, or procedural rights that are themselves harder to entrench.

**Proposition 6.25** (Re-rigging risk)**.** *Let $M'$ be a refounding of a rigged foundation $M^*$ that de-rigs but does not anti-rig. Then $M'$ is itself rigged in favor of some coalition $S'$, and by Proposition 6.12, $S'$'s advantage is self-reinforcing if $M'$ preserves the structural features of rigging.*

*Sketch.* By Proposition 6.5, $M'$ carries a legitimacy deficit. If $M'$ preserves the structural features of rigging—explicit favoritism, path dependence, information asymmetry, status-quo bias, or coordinative selection (Definition 6.10)—then the founder of $M'$ has encoded an advantage in it, just as the founder of $M^*$ did. The advantage is in favor of $S'$, the coalition that carried out the refounding. If $M'$ also preserves the procedural rights that made $M^*$ self-reinforcing, then $S'$'s advantage is self-reinforcing. □

*Remark* 6.26 (Revolution as refounding). A revolution is the limiting case of refounding: a change in $M^*$ carried out against the procedural rights of the beneficiary coalition, rather than with its consent. The framework *explains* revolutions as the de-rigging mechanism of last resort when within-rules reform is blocked. It does not *justify* any particular revolution, for two reasons: every refounding carries its own legitimacy deficit (Proposition 6.5), and every refounding risks re-rigging (Proposition 6.25). The framework gives the diagnostic for when a revolution is structurally necessary, not the normative criterion for when it is just. The latter requires a theory of justice the framework does not supply.

**Corollary 6.27** (Durability of anti-rigging)**.** *Anti-rigging is the only durable response to rigging. De-rigging without anti-rigging resets the board for the next founder (Proposition 6.25). The question facing any de-rigging coalition is therefore not only* whose advantage should be removed *but* what structure should make the next advantage harder to lay.

The founder advantage is therefore not unconditional. It persists as long as the beneficiary coalition holds the procedural rights to block reform, and it erodes when the coalition fragments, when new players accumulate countervailing leverage, or when the rules are reinterpreted by later readers. A revolution is the refounding that de-rigs when erosion is blocked; whether it succeeds in de-rigging depends on whether it also anti-rigs.

## 6.6 Domains of founding

The structure is not an analogy across domains; it is the same structural moment in different registers. In each, the foundation may be rigged, and in each, the characteristic form of rigging is different.

- **Political order.** Foundation: the constitutional settlement. Founder: the constituent power. Modes: revolution (force), constitutional convention (agreement), common-law evolution (convention), inherited settlement (accident). *Characteristic rigging:* malapportionment, gerrymandering, supermajority requirements, entrenchment of particular

  rights or interests, electoral systems that favor the founding coalition.

- **Business order.** Foundation: the incorporation—charter, cap table, governance. Founder: the entrepreneur, founding team, or investors. Modes: founder's fiat (force), partnership agreement (agreement), industry practice (convention), inherited structure (accident). *Characteristic rigging:* dual-class stock, founder shares, board composition, vesting schedules, anti-takeover provisions—all devices that encode the founder's advantage into the corporate constitution.

- **Family order.** Foundation: the household formation. Founder: the couple, the parent, the extended family. Modes: one partner's insistence (force), mutual vow (agreement), family tradition (convention), circumstance (accident). *Characteristic rigging:* inheritance rules, naming conventions, gender roles, the unspoken allocation of labor and authority that the household inherits from its own founding.

- **Personal order.** Foundation: the conditions of birth and formation—family, country, body, era, language. Founder: parents, society, circumstance, or no one. Modes: imposed circumstance (force), chosen commitment (agreement), inherited habit (convention), sheer contingency (accident). *Characteristic rigging:* the family one is born into, the country's institutions, genetic endowment, the era's opportunities. The personal foundation is the paradigm case of rigging: one did not choose it, and it shapes every subsequent allocation.

### 6.7 Consequences

(1) **Legitimacy is downstream of the foundation, and rigging amplifies the deficit.** An allocation is legitimate only if the rules producing it are legitimate, and the rules are legitimate only if the founding that established them is legitimate. Since every founding carries a deficit (Proposition 6.5), legitimacy is never fully grounded. When the foundation is rigged (Definition 6.7), the deficit is not merely abstract but visible in every allocation that follows: the disadvantaged can point to the foundation as the source of their disadvantage, and the beneficiaries cannot justify it without appealing to the very rules it established.

(2) **Reform that ignores the foundation is either naive or revolutionary.** Naive reform assumes the rules are given and proposes changes within them, unaware that the rules themselves encode an initial advantage. Revolutionary reform proposes to redo the foundation—a constitutional moment, a refounding, a new founding—and thereby to eliminate the rigging. By Proposition 6.14, the revolutionary path carries its own legitimacy deficit, but it is the only path that can in principle de-rig a rigged order.

(3) **The struggle over rules is not the deepest struggle.** The deeper contention is over who founds the rules. That contention is not itself rule-governed. It is the struggle that precedes the rules, and its outcome is the foundation on which all subsequent struggles are played. In a rigged order, this deepest struggle is asymmetric: the beneficiaries defend the foundation, the disadvantaged contest it, and the rules that would resolve the contest are the very rules in dispute.

(4) **The meta-game is nonzero-sum.** A game with a live founder advantage is never purely zero-sum (Proposition 6.18). The foundation determines the size of the pie as well as its division, and the struggle over the foundation is therefore not a struggle over shares but a struggle over the terms on which shares are possible.

(5) **De-rigging is possible but not rule-governed, and only anti-rigging is durable.** Proposition 6.14 establishes that de-rigging requires refounding. Corollary 6.27 establishes that de-rigging without anti-rigging resets the board for the next founder.

The correction of a rigged foundation is therefore not a matter of applying the right rule; it is a matter of laying a new foundation, and of designing it so that the next foundation is harder to rig.

The framework's decomposition theorem and mechanism results operate *given* $M$. They tell you what follows from the rules. They do not tell you how the rules came to be, who founded them, whether the founding is legitimate, or how foundations change. Those are the questions of first allocation, and they are prior to everything the framework formalizes. A complete theory of strategic allocation would need to supplement CSA with a theory of founding, rigging, and de-rigging; CSA's contribution is to identify exactly where that theory is required and what vocabulary—foundation, founder, mode, legitimacy deficit, initial advantage, beneficiary coalition, anti-rigging—it must use.

## 7 Applications

### 7.1 Personal decision-making

A person allocating time, attention, money, and energy across health, work, relationships, learning, and rest is solving a CSA problem with $n = 1$ (or $n > 1$ if we treat future selves as distinct players). Shares: daily allocations of time, attention, and money. Claims: choices about how to spend each hour and each dollar. Rules: habits, commitments, budgets, social obligations.

Struggle: the effort of resisting temptation, maintaining focus, and keeping commitments. Contention: competing goals for the same hours; present self versus future self over consumption versus investment; identity commitments versus opportunity.

The framework's payoff here is diagnostic. The decomposition theorem says: identify the *active* struggle network among goals. Goals that rival-contend for the same resource (e.g., deep work and social media, both claiming attention) require a **rule**—a commitment device, a schedule, a friction—not merely a preference. Goals that are cooperative (exercise and sleep) or coordinative (a morning routine) do not require willpower; they require alignment. This reframes self-control as **mechanism design on the self**, in parallel with how the empirical literature on commitment devices, implementation intentions, and choice architecture describes it.

The framework also locates the personal analogue of the first allocation. The conditions one was born into—family, country, body, era, language—are the foundation of one's personal order. They were not chosen. They cannot be fully undone. A personal foundation can be rigged too. The family one is born into, the country's institutions, one's genetic endowment, the era's opportunities—these systematically advantage some and disadvantage others, and the advantage often tracks nothing the person did. Recognizing this is not a counsel of resentment; it is a counsel of realism. One can work within a rigged foundation, and occasionally, at great cost, refound some small part of it. But one cannot pretend the rigging is not there.

### 7.2 Politics

Political competition is CSA with $M$ itself contested. Players: voters, parties, bureaucracies, interest groups, courts. Shares: budget, authority, legitimacy, agenda control, votes. Rules: constitutions, electoral systems, legislative procedure, federalism. Struggle: campaigning, lobbying, litigation, protest. Contention: rival over budget and authority;

cooperative within coalitions and parties; coordinative over constitutional conventions.

Five CSA-derived insights:

(1) **Rule contention is meta-contention.** The most consequential political struggles are over $M$—the rules that determine $\Phi$—not over a single allocation. Constitutional design, gerrymandering, and procedural reform are contests over the allocation mapping itself.

(2) **Coalition formation is cooperative contention.** Parties that cooperative-contend over shared goals but rival-contend over shares form coalitions whose stability is a core condition (Proposition 4.4).

(3) **Decomposition localizes conflict.** Many political conflicts are strategically separable across policy domains; the contention graph reveals which domains are genuinely coupled and which are artificially bundled. Rigging can create artificial coupling (§6.3), and unbundling is itself a de-rigging move.

(4) **The deepest struggle is over the foundation, and the foundation may be rigged.** Beneath rule contention lies contention over who founds the rules. Constitutional politics is different in kind from ordinary politics because it is not rule-governed in the same way: it has no referee, and its outcome is the foundation on which subsequent rule-governed politics rests (§6). When the foundation is rigged—malapportionment, gerrymandering, entrenchment of particular interests—the disadvantaged contest the foundation itself, and the beneficiaries defend it. The rules that would resolve the contest are the very rules in dispute. This is why constitutional politics is both the most consequential and the most dangerous form of political struggle.

(5) **Revolution is the de-rigging mechanism of last resort.** When rigging is self-reinforcing and within-rules reform is blocked, refounding is the only remaining path (§6.5). The framework explains revolutions as structurally necessary under these conditions; it does not justify

any particular revolution, because every refounding carries its own legitimacy deficit and risks re-rigging. The normative question requires a theory of justice that the framework does not supply.

### 7.2.1 Political alignment: left and right

The framework provides a structural account of the left–right political axis. In CSA terms, the axis is the trace, in political space, of the contention between the beneficiaries of the foundation and its disadvantaged. The two positions are not arbitrary; they are responses to different structural positions in the contention graph.

**Definition 7.1** (De-rigging and foundation-defending coalitions)**.** A *de-rigging coalition* is a coalition $D \subseteq \mathcal{N}$ whose members are disadvantaged by the foundation ($\alpha_i < 0$ for $i \in D$) and who pursue de-rigging (§6.5) as their primary political objective. A *foundation-defending coalition* is a coalition $F \subseteq \mathcal{N}$ whose members are advantaged by the foundation ($\alpha_i > 0$ for $i \in F$) and who pursue the preservation of the foundation as their primary political objective.

The framework predicts that the axis between $D$ and $F$ will be stable whenever the foundation is self-reinforcing (Definition 6.11): the beneficiaries have an interest in defending what advantages them, and the disadvantaged have an interest in contesting what disadvantages them. The axis is not a matter of opinion; it is a matter of structural position.

**Proposition 7.2** (The left–right axis)**.** *In a CSA problem with a self-reinforcing rigged foundation, the de-rigging coalition $D$ and the foundation-defending coalition $F$ have opposing preferences over the foundation $M^*$: $D$ prefers a refounding that reduces $\alpha_i$ for $i \in F$, and $F$ prefers the status quo. The axis between them is the political expression of the contention structure.*

*Sketch.* By Proposition 6.12, $F$ can block within-rules reform that reduces its advantage. By Proposition 6.14, $D$'s only path to de-rigging is refounding. Hence $D$ prefers refounding and $F$ prefers the status quo. The axis between them is the trace of this structural opposition. □

Three refinements are needed.

First, **the mapping is not identity**. Not every member of $D$ is on the left, and not every member of $F$ is on the right. The contention graph is multi-dimensional: a player can be disadvantaged by one rigging and advantaged by another. The left–right axis is a *composite* of several structural positions, and the composite is stable because the positions tend to correlate, not because they are identical.

Second, **the zero-sum disagreement is itself a move in the over-rules game**. The left characteristically claims that the game is not zero-sum (de-rigging can make everyone better off), and the right characteristically claims that it is (redistribution is a transfer). By Proposition 6.18, the left is correct if the founder advantage is live, and the right is correct if it is fixed. The disagreement about whether the game is zero-sum is therefore a disagreement about whether the foundation is contestable—and that disagreement is itself a strategic move. The beneficiaries have an interest in treating the advantage as fixed (it closes the over-rules game); the disadvantaged have an interest in treating it as live (it opens the over-rules game).

Third, **the framework does not adjudicate the normative question**. It can say that both positions have a structural basis, that the left is right that foundations are often rigged, and that the right is right that refounding is dangerous and can re-rig. The synthesis is anti-rigging (§6.5): change the structure so that future rigging is harder. This is neither purely left nor purely right; it is a structural design question. But the framework cannot say how much rigging is acceptable, given the costs of de-rigging. That requires a theory of justice the framework does not supply.

*Remark* 7.3 (Mode of founding and political temperature). The mode of founding (§6) shapes the character of the left–right conflict. A foundation laid by force produces a left that treats the founding as illegitimate and a right that defends it as established. A foundation laid by agreement produces a left that argues the agreement was incomplete and a right that argues it was the best available. A foundation laid by convention produces a left that argues it was never chosen and a right that argues it has stood the test of time. A foundation laid by accident produces a conflict that is less about the founding and more about what to do next.

*Remark* 7.4 (The axis and decomposition). The decomposition theorem (§5) explains why not every issue is political. An issue becomes political when it touches the foundation—when the allocation of shares is coupled to the rules that determine them. Technical, administrative, and coordinative issues are not political in this sense, and the framework predicts that they will be settled by expertise or convention rather than by left–right contest. The left–right axis appears on issues where the foundation is genuinely at stake.

### 7.2.2 The space of political axes

The framework explains not only the left–right axis but the full space of political axes. The space is not arbitrary: it is isomorphic to the space of contestable parameters of the CSA problem.

**Proposition 7.5** (Axis typology)**.** *Every political axis is a disagreement about one of five parameters of the CSA problem:*

*(i) the* object *of allocation—which shares are contended ($\mathcal{R}$);*

*(ii) the* boundary *of the player set—who is a player ($\mathcal{N}$);*

*(iii) the* scope *of the rules—what the rules cover ($M$);*

*(iv) the* legitimacy *of the foundation—whether the founding is justified ($M^*$); or*

*(v) the* mode *of change—how the order is to be changed.*

*Individual axes may combine elements of multiple families, but each has a primary family.*

*Sketch.* The CSA problem is a tuple $(\mathcal{N}, \mathcal{R}, \Theta, A, I, B, \Phi, M, M^*, \text{mode}, \text{level})$. Each parameter can be fixed or contested. When a parameter is contested, a political axis emerges: a stable disagreement about the parameter's value. The five families correspond to the five parameters that generate stable axes. Method axes are the composite of mode and level. The typology is exhaustive because every disagreement about a CSA problem must be about one of its parameters. □

| **Family** | **Parameter** | **Axes** | **Question** |
|---|---|---|---|
| Object | $\mathcal{R}$ | Left–right (economic), materialist–postmaterialist, green–growth | What is divided? |
| Boundary | $\mathcal{N}$ | Globalist–nationalist, hawk–dove, urban–rural, young–old | Who is in the game? |
| Rule-scope | $M$ | Libertarian–authoritarian, federal–unitary | How much is governed? |
| Foundation-legitimacy | $M^*$ | Progressive–conservative, populist–elitist, religious–secular | Is the founding justified? |
| Method | mode | Reform–revolution, moderate–radical | How should it change? |

Each family has a distinct structural logic. We treat each in turn.

**Object axes.** Left–right (economic) has been analyzed in §7.2.1. Materialist–postmaterialist [7] is a disagreement about which shares matter: as material shares become secure, the resource set $\mathcal{R}$ expands to include identity, autonomy, and recognition. Green–growth extends $\mathcal{R}$ to environmental goods and future generations. The green axis is structurally distinctive because the expansion of $\mathcal{R}$ brings in players who cannot contest—future generations and non-human nature. It requires *trustee* mechanisms, because the affected parties are voiceless.

**Boundary axes.** Globalist–nationalist is a disagreement about $\mathcal{N}$: who is inside the order? A nationalist says the foundation is for citizens; a globalist says it is for all affected parties. Expanding $\mathcal{N}$ is structurally a refounding (§6), because the foundation was laid for a smaller set—which is why the axis is so bitter. Hawk–dove is a disagreement about whether other states are rival or cooperative contenders, and the nonzero-sum theorem (§6.4) applies directly: if the international foundation is live, doves are right; if it is fixed, hawks may be. Urban–rural and young–old are structural positions in the contention graph rather than disagreements about a parameter: they reflect different resource bases and different time horizons.

**Rule-scope axes.** Libertarian–authoritarian is a disagreement about the domain of the allocation mapping $\Phi$: whether it is near-identity on private claims (libertarian) or a thick collective allocation (authoritarian). Federal–unitary is a disagreement about the level at which rules are made. The decomposition theorem (§5) applies to both: decentralization adds value when issues are separable, centralization when they are coupled.

**Foundation-legitimacy axes.** Progressive–conservative is structurally identical to economic left–right, but over cultural shares: the progressive sees the cultural foundation as rigged and wants to refound it; the conservative sees it as established and wants to preserve it. This is why the two axes correlate: they are the same structure over different shares. Populist–elitist is a *re-rigging* axis: the populist claims the foundation has been captured by a new elite; the elitist defends the current coalition. By Proposition 6.25, populist refoundings will re-rig rather than anti-rig, because populism is a de-rigging movement without an anti-rigging program. Religious–secular is the most structurally distinctive: it is a disagreement about the mode of founding, but a special one, because the religious view treats the founding as outside the four modes—as divine ordination rather than force, agreement, convention, or accident. The framework cannot adjudicate, because it has no category for transcendent authority. It can only note that the religious view refuses the regress, while the secular view lives with it.

**Method axes.** Reform–revolution is a disagreement about whether the rigging is self-reinforcing: if it is, reform is blocked and refounding is the only path (Proposition 6.14); if it is not, reform is sufficient and refounding is dangerous. Moderate–radical is a disagreement about the coupling of issues: moderate when issues are separable, radical when they are coupled. The decomposition theorem applies to both.

*Remark* 7.6 (The correlation of axes). The axes correlate because the layers correlate: the beneficiaries of a rigged economic foundation tend to defend the cultural foundation, prefer the within-rules game, and see the founding as legitimate. But the correlation is not identity: a person can be economically left and culturally conservative, or vice versa. The composite is stable because the positions tend to correlate, not because they are identical.

*Remark* 7.7 (New axes). The framework predicts that new axes will emerge as new resources become scarce. As data, attention, and identity become scarce, new object axes emerge around them (digital rights, AI governance, identity politics). The framework predicts these will fit the same structural pattern: de-rigging vs. foundation-defending over the new resource.

### 7.3 Business strategy

A firm allocating capital, capacity, talent, data, and attention across markets and functions, while rivals, suppliers, and regulators do the same, is solving a CSA problem with $M$ partly chosen by the firm (contracts, org design) and partly imposed (regulation, antitrust). Shares: market share, margin, talent, data, customer attention. Claims: investment, hiring, pricing, product moves. Rules: contracts, auctions, market structure, regulation. Struggle: price competition, advertising spend, litigation, lobbying. Contention: rival over customer attention and capacity; cooperative over complementary assets; coordinative over standards.

Five CSA-derived strategic questions:

(1) **Which resources are rival-contended?** Rivalry over customer attention (advertising), talent (hiring), and capacity (price wars) generates different strategic dynamics than cooperative contention over complementary assets (alliances, ecosystems). The contention graph tells you which game you are in.

(2) **What is the allocation mapping?** Strategy is often the design of $\Phi$ itself—platform rules, pricing mechanisms, supply contracts—not just a choice of $a$ within fixed $\Phi$.

(3) **When does strategy decompose?** Multi-business firms can often be decomposed into strategically separable units. The decomposition theorem tells you when central coordination adds value (coupled contention) and when it adds only overhead (separable contention).

(4) **What is the founding, and is it rigged?** The incorporation—the charter, the cap table, the governance—is the first allocation of the firm (§6). Founder shares, dual-class stock, board composition, and anti-takeover provisions are devices that encode the founder's advantage into the corporate constitution. Founder disputes are bitter because they are about the foundation, not about ordinary management. Governance reforms are contested for the same reason, and they often fail because the rigged foundation gives the beneficiaries the procedural rights to block de-rigging. "Who decides" is a deeper question than "what is decided," and "who decided the founding" is deeper still.

(5) **Is the competitive game zero-sum?** Rarely. A firm that mistakes a live founder advantage for a fixed pie—its own or a rival's—will misread the strategic situation (§6.4). Markets are rival at the margin but nonzero-sum over their foundations: regulation, standards, market structure, and technology shift the total, not merely its division. A firm that invests in shaping these foundations is playing the over-rules game; one that only optimizes within them is playing the

within-rules game. Both can be correct, but confusing the two is a strategic error.

## 8 Discussion

### 8.1 Scope

CSA is a *modeling language*, not a theory of everything. It is useful exactly when allocation, contention, and struggle are the salient features of a situation—which is most of the time, but not always. Pure signaling games, pure epistemic puzzles, and games whose interest lies in equilibrium refinement rather than allocation are better described in standard terms.

### 8.2 Bounded rationality

CSA can accommodate bounded rationality by restricting $A_i$ (limited claims), $I_i$ (limited information), $B_i$ (limited struggle capacity), or the strategy space (limited contingent plans). But it does not *explain* bounded rationality; that requires a theory of cognition, attention, and habit that CSA does not supply.

## 8.3 Norms and informal rules

The rule structure $M$ is deliberately broad. It includes formal rules—contracts, laws, constitutions—and informal ones—norms, conventions, expectations. This is important because many of the most consequential rules in personal, political, and business life are informal. CSA does not model how norms form or change; it treats them as part of $M$ and asks what difference they make to allocation and equilibrium. The formation of norms is a special case of the problem of first allocation (§6): a norm is a rule whose founding is often conventional or accidental rather than deliberate.

## 8.4 The first allocation

The framework brackets the problem of first allocation (§6): it takes $M$ as given or chosen, and derives allocations from it. This is a genuine limit, not a minor one. The legitimacy of any allocation depends on the legitimacy of the rules, and the legitimacy of the rules depends on the founding. A complete theory would need to say how first allocations are made, justified, and changed. CSA does not supply that theory; it identifies where it is needed and provides the vocabulary—foundation, founder, mode of founding, legitimacy deficit, initial advantage, beneficiary coalition—in which the theory can be stated.

The theory of the rigged foundation (§6.3–§6.5) sharpens the limit. It shows that the first allocation can systematically advantage some players, that within-rules reform cannot eliminate this advantage when the rigging is self-reinforcing, and that de-rigging requires refounding—which carries its own legitimacy deficit and risks re-rigging. The framework therefore identifies not only where a theory of founding is needed but also why it is hard: the correction of a rigged foundation is itself a founding, and foundings are not rule-governed. And it shows that the meta-game is nonzero-sum, which is why zero-sum reasoning—though correct within a fixed foundation—is systematically misleading about the situations that matter most.

## 8.5 Normative questions

CSA is descriptive and analytical, not normative. It can identify Pareto-inefficient allocations, contention-driven conflict, and the welfare consequences of alternative rules. It cannot by itself say what allocation is *just*. That requires an ethical theory, which CSA can inform but not replace. The problem of first allocation sharpens this point: the legitimacy deficit of every founding (§6) means that no purely procedural theory of justice can fully ground itself. Normative theory must confront the founding, not merely the rules that follow from it. And the theory of the rigged foundation (§6.3) adds that de-rigging is not a rule-governed correction but a refounding—which means that normative theory must also confront the conditions under which a refounding is justified, and this is a question the framework cannot answer.

The analysis of political axes (§7.2.1–§7.2.2) has a further implication. The framework shows that the left–right axis, and the other axes, are structural responses to the contention between beneficiaries and disadvantaged. It follows that no political position can be refuted merely by showing that it reflects a structural interest. Every position reflects a structural interest; the question is whether the position is also correct. The framework cannot answer that question, but it can sharpen it: it can show what each axis is about, and what a resolution would require.

## 8.6 Relationship to standard game theory

CSA adds no expressive power beyond extensive-form games: any CSA problem can be encoded as an extensive-form game by treating $M$ and $\Phi$ as part of the game tree, and every extensive-form game is a CSA problem. The contribution is not a new theorem but a new **primitive ontology**—one that treats shares, contention, struggle, rules, founders, foundations, and axes as first-class objects. Six organizational claims follow:

(1) **Allocation is the outcome of interest.** Games are often analyzed as if the payoff were the primitive; CSA insists that the share is the primitive and payoff is a function of it. This matters for welfare analysis, fairness, and mechanism design.

(2) **Rules are explicit.** Treating $M$ as an object—sometimes exogenous, sometimes contested, sometimes designed—clarifies the difference between playing a game and choosing one.

(3) **Contention and struggle are structural.** Making contention a graph and struggle a weighted network, rather than implicit properties of payoffs, permits decomposition, diagnosis, and targeted intervention.

(4) **Foundations are explicit.** Naming the problem of first allocation clarifies the boundary of the framework and the point at which a theory of founding must take over.

(5) **Rigging is structural.** Naming the rigged foundation clarifies why some games are not zero-sum, why within-rules reform so often fails, and why de-rigging requires refounding with anti-rigging if it is to be durable.

(6) **Axes are structural.** Deriving political axes from the contestable parameters of the CSA problem clarifies why the axes exist, why they correlate, and why they persist across changes in the specific issues.

In practice, the choice of primitive ontology matters: it shapes which questions seem natural, which comparative statics are visible, and which interventions are conceivable. CSA is offered in that spirit.

# 9 Conclusion

Games are struggles over shares, under rules, where what each player gets depends on what everyone claims. Making that idea precise yields a framework—Contested Strategic Allocation—that unifies normal-form and extensive-form games, exchange economies, cooperative games, mechanism design, and single-agent planning as instances of a single structure. The framework's central structural object is the contention graph, which determines when a problem decomposes, where strategic difficulty lies, and what mechanisms can do; its central process object is the struggle network, which determines how contention is pursued and at what cost; its central limit is the problem of first allocation, which marks the point at which the framework falls silent; its central consequence is that the foundation is not neutral: it may be rigged, and when it is, the rigging is self-reinforcing, the meta-game is nonzero-sum, and de-rigging requires refounding with anti-rigging if it is to be durable; and its central application to politics is the derivation of the left–right axis and the full space of political axes from the contention structure.

In personal life, the struggle network reveals which goals require commitment devices rather than willpower, and the first allocation marks the conditions one did not choose. In politics, it reveals which conflicts are genuinely coupled and which are artificially bundled, why rule contests dominate, why the deepest struggles are over the founding, why revolutions are the de-rigging mechanism of last resort, and why the left and the right are structural responses to the same contention. In business, it reveals which resources generate rivalry, which generate cooperation, when central coordination adds value, why founder disputes are bitter, and why the competitive game is rarely the fixed pie it appears to be.

The framework does not solve these problems. It organizes them. And it is honest about where its organization ends: at the foundation, which no rule of the order can justify, and which every founder—conqueror, constitution-writer, entrepreneur, parent—must lay without the justification that only the rules themselves could provide. The foundation may be rigged, and when it is, the rigging is self-reinforcing: within-rules reform preserves it, and de-rigging

requires a new founding with its own legitimacy deficit. This is the deepest thing the framework can say, and the last: **the game is played on a field that someone laid, and the laying of the field is not itself a move in the game.** The field can be relaid, but a relaying that does not change the *terms* of laying merely resets the board for the next founder. The deepest lesson of the first allocation is not that the game is rigged but that rigging is a structure—and structures can be redesigned, at the cost of a new founding with its own legitimacy deficit.

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# A Summary of Notation

| Symbol | Ordinary-language gloss | Formal meaning |
|---|---|---|
| $\mathcal{N}$ | Players | Set of players |
| $\mathcal{R}$ | Resources | Set of resource types |
| $\Theta$ | States of the world | State space |
| $A_i$ | Claims | Player $i$'s claim set |
| $I_i$ | What $i$ knows | Information partition |
| $B_i$ | Struggle capacity | Struggle budget (scalar) |
| $F_i$ | Which efforts are feasible | Effort set |
| $\mathcal{X}$ | Shares | Allocation space |
| $\Phi$ | How claims and struggle become shares | Allocation mapping |
| $M$ | Rules | Rule structure |
| $M^*$ | Foundation | First allocation (regress terminus) |
| $\alpha_i$ | Initial advantage | Advantage conferred by the foundation |
| $u_i$ | Player $i$'s payoff | Payoff function |
| $T$ | How things evolve | Transition map |
| $\sigma_i$ | Plan | Strategy |
| $G_r^{\text{riv}}, G_r^{\text{coop}}, G_r^{\text{coord}}$ | Who contends with whom | Contention graphs |
| $\tilde{G}_r$ | How hard they struggle | Weighted struggle network |
| $\kappa(\mathcal{G})$ | Density of rivalry | Rival contention index |
| $\sigma(\mathcal{G})$ | Intensity of struggle | Struggle intensity |